# BOUNDED SUMS VERSUS BOUNDED DIFFERENCES: A NOTE ON ERDŐS PROBLEM #863

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Abstract. For an integer $r \geq 1$, let $S_r(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ in which every integer has at most $r$ representations $a + a'$ with $a, a' \in A$, $a \leq a'$, and let $D_r(N)$ be the largest size of a set $B \subseteq \{1, \dots, N\}$ in which every positive integer has at most $r$ representations $b - b'$ with $b, b' \in B$. Erdős, recalling conversations with D. Berend and R. Freud, asked whether, if the asymptotic constants in $S_r(N) \sim c_r\sqrt{N}$ and $D_r(N) \sim c'_r\sqrt{N}$ exist, one has $c'_r < c_r$ for every integer $r \geq 2$. We observe that this follows from results in the literature: the Erdős–Turán argument for Sidon sets gives $D_r(N) \leq (\sqrt{r} + o(1))\sqrt{N}$ (indeed $D_r(N) \sim \sqrt{rN}$ was already known), while a construction of Cilleruelo, Ruzsa and Trujillo gives $S_r(N) \geq \big((r+s)/\sqrt{r+2s} - o(1)\big)\sqrt{N}$ with $s = \lfloor r/2 \rfloor$, and $(r+s)/\sqrt{r+2s} > \sqrt{r}$. Taking Singer's theorem and the prime number theorem as standard inputs, we give complete proofs of the two bounds needed for the separation, and survey what is known about the two constants.

**About this version.** This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from** `https://boonsuan.github.io/erdos863_old.pdf`**.** The mathematics is that of the original note, except that the upper bound is now proved with an explicit error term and the lifting lemma is stated, as in the work of Cilleruelo, Ruzsa and Trujillo, for arbitrary layer patterns. This version adds figures, a worked example, a survey of the known bounds, and corrected attributions; in particular, the asymptotic formula $D_r(N) \sim \sqrt{rN}$ was already in the literature.

## 1. Introduction

1.1. **The problem.** In his 1992 paper *Some of my forgotten problems in number theory*, Erdős recalls a question that he and D. Berend had noticed during a visit to Ben-Gurion University in Beer Sheva, and that he later reformulated independently with R. Freud before remembering the earlier conversation [Er92, pp. 39–40]. In the notation of the problem collection [Blo] it reads as follows.

**Problem** (Erdős Problem #863)**.** *Let $r \geq 2$. Let $A \subseteq \{1, \dots, N\}$ be a set of maximum cardinality such that for every $n$ there are at most $r$ solutions of $n = a + b$ with $a, b \in A$, $a \leq b$, and let $B \subseteq \{1, \dots, N\}$ be a set of maximum cardinality such that for every $n \geq 1$ there are at most $r$ solutions of $n = a - b$ with $a, b \in B$. If $|A| \sim c_r N^{1/2}$ and $|B| \sim c'_r N^{1/2}$, is it true that $c_r \neq c'_r$? Is it true that $c'_r < c_r$?*

Sets of the first kind are the $B_2[r]$ *sets* (or generalized Sidon sets); sets of the second kind have been called $r$*-Golomb rulers* or $B_2^-[r]$ sets [CMT15]. For $r = 1$ the two conditions coincide: both say that $A$ is a *Sidon set*, that is, all sums $a + a'$ with $a \leq a'$ are distinct, or equivalently all positive differences are distinct. So $c_1 = c'_1$, and the theorem of Erdős and Turán [ET41] (an upper bound $\sqrt{N} + O(N^{1/4})$) together with Singer's difference sets [Sin38] (whose use for this

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purpose Erdős credits to Chowla and himself [Er92, p. 37]) gives $c_1 = c_1' = 1$. For $r \geq 2$ the conditions are different, and Erdős expected sums to allow larger sets. As evidence he quoted two results from his Hungarian paper with Freud [EF91]: if a single integer is allowed more than one representation as a sum, one can take about $\frac{2}{\sqrt{3}}\sqrt{N}$ elements, whereas the corresponding relaxation for differences still forces $(1+o(1))\sqrt{N}$. In Erdős's words these results "seem to give hope" for $c_r' < c_r$ [Er92, p. 40].

1.2. **The answer.** Throughout, $r$ and $N$ are positive integers, and $r$ is fixed as $N \to \infty$ unless otherwise indicated. Write $[N] = \{1, \dots, N\}$. For a finite set $A \subseteq \mathbb{Z}$ and $m \in \mathbb{Z}$ let

$$R_A(m) = \#\{(a, a') \in A^2 : \ a \leq a', \ a + a' = m\}$$

be the number of *unordered* representations of $m$ as a sum of two elements of $A$ (repetition $a = a'$ allowed), and for a finite set $B \subseteq \mathbb{Z}$ and an integer $d \geq 1$ let

$$\Delta_B(d) = \#\{(b, b') \in B^2 : \ b - b' = d\}.$$

The two maxima of the problem are[1]

$$S_r(N) = \max\{|A| : \ A \subseteq [N], \ R_A(m) \leq r \text{ for all } m \in \mathbb{Z}\},$$
$$D_r(N) = \max\{|B| : \ B \subseteq [N], \ \Delta_B(d) \leq r \text{ for all } d \geq 1\}.$$

Sums are counted without order, as for Sidon sets: with ordered counting a pair of distinct elements would contribute two ordered representations, so for $r = 1$ an admissible set would have at most one element. Only positive differences are restricted ($0 = b - b$ has $|B|$ representations), and a positive difference $b - b'$ has $b > b'$, so for differences there is no ordered/unordered distinction to make. These are the conventions of the problem page [Blo]. Both maxima are nondecreasing in $N$: every admissible subset of $[N]$ remains admissible in $[N+1]$.

**Theorem 1.1.** *Let $r \geq 2$ be an integer and $s = \lfloor r/2 \rfloor$. Then*

$$\limsup_{N\to\infty} \frac{D_r(N)}{\sqrt{N}} \ \leq \ \sqrt{r} \ < \ \frac{r+s}{\sqrt{r+2s}} \ \leq \ \liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}}.$$

The middle inequality is elementary: since $s \geq 1$ when $r \geq 2$,

$$\Big(\frac{r+s}{\sqrt{r+2s}}\Big)^2 - r = \frac{(r+s)^2 - r(r+2s)}{r+2s} = \frac{s^2}{r+2s} > 0. \tag{1}$$

So only the two outer bounds need proof. Erdős wrote the maxima as $(c_r + o(1))\sqrt{n}$ and $(c_r' + o(1))\sqrt{n}$, presupposing that the limits exist; Theorem 1.1 does not need this, and shows in particular that *if* both limits exist then $c_r' \leq \sqrt{r} < c_r$. So the answer to both of Erdős's questions is yes.

More is true. It is known that $D_r(N) \sim \sqrt{rN}$ [CMT15, ST25] (see Section 4), so $c_r'$ exists and equals $\sqrt{r}$, and Theorem 1.1 says that every limit point of $S_r(N)/\sqrt{N}$ is strictly larger than $c_r'$. Whether $c_r$ itself exists is open for every $r \geq 2$ [CRV10, Pla01]. For $r = 2$, for instance, $c_2' = \sqrt{2} \approx 1.414$, while the best lower bound for $S_2(N)/\sqrt{N}$ that we know of is $4/\sqrt{7} \approx 1.512$ [HP02] and the best upper bound we know of is 2.2851 [HP18].

[1] Erdős takes $b_1 < \cdots < b_\ell < n$, which replaces $D_r(n)$ by $D_r(n-1)$; this does not affect any asymptotic statement.

1.3. **The idea of the proofs.** *Differences* (Section 2). Slide a window of length $H$, with $\sqrt{N} \ll H \ll N$, across $\{1, \dots, N\}$. Two elements of $B$ in a common window differ by less than $H$, and there are only $H-1$ such differences, each available at most $r$ times. Counting pairs of elements sharing a window from below (by Cauchy–Schwarz) and from above (by the multiplicity bound) shows that $B$ cannot have average density more than about $\sqrt{r/N}$; see Fig. 1.

*Sums* (Section 3). By contrast, pairs lying in short windows can have sums spread across the full interval of possible sums, so no such local count is available, and the construction exploits this. Take a Singer set $C$ modulo $Q = q^2+q+1$, a set of $q+1$ residues whose nonzero differences are all distinct, and place copies of it in several consecutive blocks ("layers") of length $Q$, using only the layers $u$ in a fixed finite pattern $P$. Because $C$ is a Sidon set modulo $Q$, a sum determines the unordered pair of residues of its two summands, and the representation count of the lifted set reduces to the number of ways of writing a layer index as $u+v$ with $u, v \in P$, *counted with order*: when the two residues differ, the summand with the first residue may sit in either of the two layers. A pattern $P \subseteq \{0, \dots, K-1\}$ with at most $r$ such ordered representations yields $S_r(N) \gtrsim (|P|/\sqrt{K})\sqrt{N}$. An interval $P = \{0, \dots, r-1\}$ gives only $\sqrt{r}$; adding a second block of $s$ layers after a gap, as Cilleruelo, Ruzsa and Trujillo did, gives $(r+s)/\sqrt{r+2s}$ (Figs. 2 and 3).

1.4. **A heuristic.** Comparing expected counts suggests that the sum condition may permit denser sets. For a random set $X \subseteq [N]$ of density $\delta$, a positive difference $d$ has about $\delta^2(N-d)$ representations, so the busiest differences, the short ones, have about $\delta^2 N$ each. A sum $m$ has about $\delta^2 \min(m, 2N-m)$ ordered representations, so the busiest sums, those near $N$, also have about $\delta^2 N$ ordered representations, but only about $\delta^2 N/2$ unordered ones: $a+a'$ and $a'+a$ are the same sum, while $b-b'$ and $b'-b$ are different differences. This is only a qualitative guide. Expected counts do not control the maximum count (for $r=1$ the two problems have the same answer), and a uniform density profile does not predict the optimal constant: for large $r$ the ratio of the two constants is governed by a convolution optimization (Section 4.2), not by the factor $\sqrt{2}$ that this calculation suggests. For fixed small $r$ one needs explicit constructions, and the one used here gains a factor of only about 1.03–1.06 (Table 1).

1.5. **Sources.** Both halves of Theorem 1.1 are in the literature. The upper bound for differences is the Erdős–Turán argument [ET41], which goes through unchanged for general $r$; Caicedo, Martos and Trujillo prove the explicit bound $D_r(N) \le \sqrt{rN} + (rN)^{1/4} + 1$ [CMT15, Thm. 2.5] (which they attribute to earlier work of Trujillo, García and Velásquez) and the matching lower bound, so that $D_r(N) \sim \sqrt{rN}$ [CMT15, Thm. 3.10]; Schmutz and Tait [ST25, Thm. 3] later proved the same asymptotic formula independently. The lower bound for sums is Theorem 2.1 of Cilleruelo, Ruzsa and Trujillo [CRT02]. Their proof lifts a Singer difference set [Sin38] along an arbitrary layer pattern with bounded ordered representation counts [CRT02, Lemma 2.2], and Plagne's survey [Pla01, §2.2, (10)–(11)] states this principle for general $B_h[g]$ sets and optimizes over patterns. The observation that these two results settle Problem #863 was made by Boon Suan Ho with GPT-5.4 Pro in April 2026 [Ho26, Thr]; the problem page [Blo] now records it.

## 2. Differences: the Erdős–Turán argument

The following is the Erdős–Turán argument [ET41], in a form with overlapping windows; for $r=1$ it is their bound $\sqrt{N} + O(N^{1/4})$ for Sidon sets. (See O'Bryant's survey [OB04] for

the many later refinements of this argument for Sidon sets, and [CMT15, ST25] for two other proofs of the same bound for general $r$.)

**Proposition 2.1.** *For all integers $r, N \geq 1$,*

$$D_r(N) < \sqrt{rN} + \Big(\frac{\sqrt{r}}{2} + 2\Big)N^{1/4}.$$

*In particular* $\limsup_{N\to\infty} D_r(N)/\sqrt{N} \leq \sqrt{r}$.

*Proof.* Let $B \subseteq [N]$ satisfy $\Delta_B(d) \leq r$ for all $d \geq 1$, and put $M = |B|$.

*A crude bound.* The $\binom{M}{2}$ pairs $b > b'$ in $B$ have differences in $\{1, \dots, N-1\}$, each value occurring at most $r$ times, so $M(M-1) \leq 2r(N-1) < 2rN$. This forces $M \leq 2\sqrt{rN}$: otherwise $M(M-1) > 2\sqrt{rN}\,(2\sqrt{rN} - 1) = 4rN - 2\sqrt{rN} \geq 2rN$, since $rN \geq 1$.

*Windows.* Fix an integer $H$ with $1 \leq H \leq N$. For each integer $x$ in the range $1 - H \leq x \leq N - 1$ (there are $N + H - 1$ of them) let

$$I_x = \{x+1, x+2, \dots, x+H\}, \qquad w_x = |B \cap I_x|.$$

An element $b \in [N]$ lies in $I_x$ exactly when $b - H \leq x \leq b - 1$, that is, in exactly $H$ of the windows. Hence $\sum_x w_x = MH$, and by the Cauchy–Schwarz inequality

$$\sum_x w_x(w_x - 1) = \sum_x w_x^2 - \sum_x w_x \;\geq\; \frac{M^2H^2}{N + H - 1} - MH. \tag{2}$$

*Counting pairs in a window.* The left side of (2) counts triples $(x, b, b')$ with $b \neq b'$ both in $I_x$. If $b > b'$ and $d = b - b'$, then both lie in $I_x$ exactly when $b - H \leq x \leq b' - 1$; this happens for $H - d$ values of $x$ if $d < H$, and for none if $d \geq H$ (see Fig. 1). Counting both orders of each pair,

$$\sum_x w_x(w_x - 1) = 2\sum_{d=1}^{H-1} (H - d)\,\Delta_B(d) \;\leq\; 2r\sum_{d=1}^{H-1}(H - d) = rH(H-1). \tag{3}$$

*Conclusion.* Combining (2) and (3) and multiplying by $(N + H - 1)/H^2$ gives

$$M^2 \;\leq\; r\Big(1 - \frac{1}{H}\Big)(N + H - 1) + M\,\frac{N + H - 1}{H}. \tag{4}$$

Since $M = O_r(\sqrt{N})$, the two error terms in (4) are $O_r(H)$ and $O_r(N^{3/2}/H)$; choosing $H$ of order $N^{3/4}$ balances them, and after taking a square root leaves an error $O_r(N^{1/4})$. Explicitly, take $H = \lceil N^{3/4} \rceil$, so that $N^{3/4} \leq H < N^{3/4} + 1$ and $H \leq N$. The first term of (4) is less than $r(N + N^{3/4})$. In the second, $(N + H - 1)/H < 2N/H \leq 2N^{1/4}$, and $M \leq 2\sqrt{rN}$, so the second term is at most $4\sqrt{r}\,N^{3/4}$. Therefore

$$M^2 < rN + (r + 4\sqrt{r})N^{3/4},$$

and since $\sqrt{a+b} \leq \sqrt{a} + b/(2\sqrt{a})$ for $a > 0$, $b \geq 0$,

$$M < \sqrt{rN} + \frac{(r + 4\sqrt{r})N^{3/4}}{2\sqrt{rN}} = \sqrt{rN} + \Big(\frac{\sqrt{r}}{2} + 2\Big)N^{1/4}. \qquad \square$$

*Remark* 2.2. Only differences $1 \leq d < \lceil N^{3/4} \rceil$ enter (3); the long differences are used only in the crude bound $M \leq 2\sqrt{rN}$. This is the precise sense in which the difference problem is *local*: a set with few repeated short differences cannot be denser than $\sqrt{r/N}$ on average. The explicit error term in Proposition 2.1 is simply a convenient bound obtained by this proof; better ones are known. Lindström [Lin69] proved $\sqrt{N} + N^{1/4} + 1$ for Sidon sets, and [CMT15, Thm. 2.5] gives $\sqrt{rN} + (rN)^{1/4} + 1$ for general $r$.

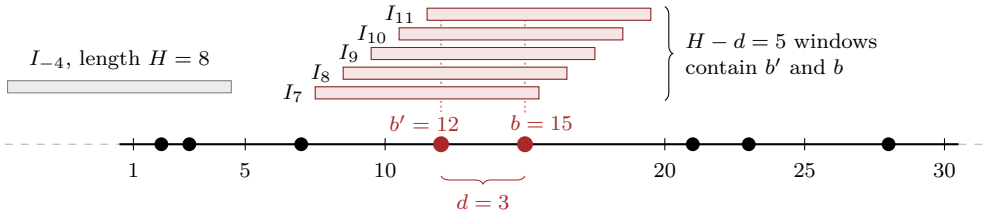



**Figure 1.** The window count in the proof of Proposition 2.1, with $N = 30$, $H = 8$ and $B = \{2, 3, 7, 12, 15, 21, 23, 28\}$, a set in which every positive difference occurs at most 3 times. The windows $I_x = \{x + 1, \dots, x + H\}$, $1 - H \le x \le N - 1$, may stick out of $[N]$, so that every element of $B$ lies in exactly $H$ of them. Two elements at distance $d < H$ lie together in exactly $H - d$ windows (here $I_7, \dots, I_{11}$). The identity (3) weights each pair at distance $d < H$ by $H - d$.

## 3. Sums: lifting a Singer set

3.1. **Cyclic Sidon sets.** The building block is Singer's construction.

**Theorem 3.1** (Singer [Sin38])**.** *Let $q$ be a prime power and $Q = q^2 + q + 1$. There is a set $C \subseteq \mathbb{Z}/Q\mathbb{Z}$ with $|C| = q + 1$ such that every nonzero residue modulo $Q$ has exactly one representation $c - c'$ with $c, c' \in C$.*

A set with this difference property is called a *planar* (or perfect) difference set. In Singer's construction, its translates correspond to the lines of the projective plane $\mathrm{PG}(2, q)$. For example, $\{0, 1, 3\}$ modulo 7 and $\{0, 1, 3, 9\}$ modulo 13 are planar difference sets. Bose [Bos42] gave a similar construction of $q$ residues modulo $q^2 - 1$ with distinct differences, which would serve equally well below. The only property we need is that the nonzero differences are distinct, which makes $C$ a Sidon set for sums modulo $Q$:

**Lemma 3.2.** *Let $C \subseteq \mathbb{Z}/Q\mathbb{Z}$ be a set whose nonzero differences $c - c'$ $(c \ne c')$ are pairwise distinct. If $c_1 + c_2 = c_3 + c_4$ in $\mathbb{Z}/Q\mathbb{Z}$ with all $c_i \in C$, then $\{c_1, c_2\} = \{c_3, c_4\}$ as multisets.*

*Proof.* If $c_1 = c_3$ then $c_2 = c_4$. Otherwise $c_1 - c_3 = c_4 - c_2$ is a nonzero difference represented by the pairs $(c_1, c_3)$ and $(c_4, c_2)$, so these pairs coincide: $c_1 = c_4$ and $c_3 = c_2$. □

3.2. **Lifting along a layer pattern.** For a finite set $P$ of nonnegative integers (the *layer pattern*) and $t \in \mathbb{Z}$, write

$$\rho_P(t) = \#\{(u, v) \in P^2 : \ u + v = t\}$$

for the number of *ordered* representations of $t$ as a sum of two elements of $P$. Order matters because, when the two residues of a sum are distinct, assigning layer $u$ to the first residue and layer $v$ to the second usually gives a different unordered pair of integers from the assignment with the layers exchanged (see Example 3.4). The following lemma is [CRT02, Lemma 2.2], with the hypothesis on $C$ phrased through differences.

**Lemma 3.3** (Lifting)**.** *Let $P \subseteq \{0, 1, \dots, K - 1\}$ satisfy $\rho_P(t) \le r$ for all $t$, and let $C \subseteq \{0, 1, \dots, Q - 1\}$ be a set of distinct residues modulo $Q$ whose nonzero differences are distinct modulo $Q$. Then the set*

$$A = \{uQ + c + 1 : \ u \in P, \ c \in C\}$$

*satisfies $A \subseteq [KQ]$, $|A| = |P|\,|C|$, and $R_A(m) \le r$ for every integer $m$.*

*Proof.* Every $x \in A$ has a unique expression $x = uQ + c + 1$ with $u \in P$, $c \in C$: $u$ and $c$ are the quotient and remainder of $x - 1$ on division by $Q$. We call $u$ the *layer* and $c$ the *residue* of $x$. This gives $|A| = |P||C|$, and $1 \le x \le (K - 1)Q + (Q - 1) + 1 = KQ$.

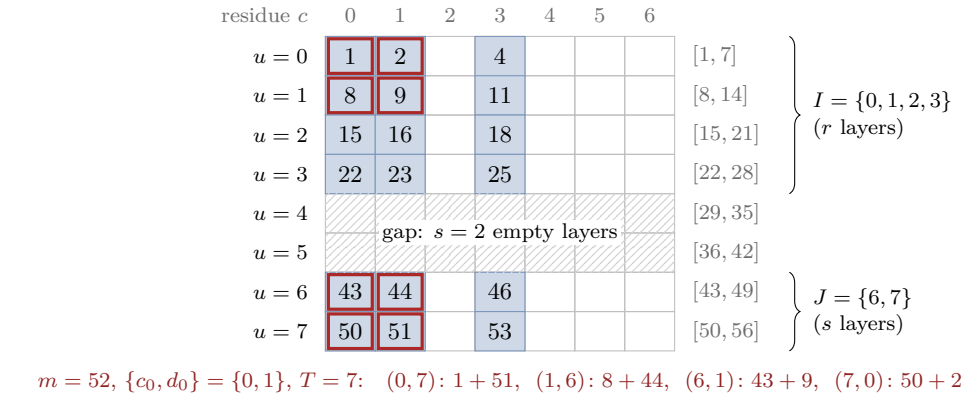



**Figure 2.** The lifted set of Lemma 3.3 for $r=4$ (so $s=2$, $P=P_4=\{0,1,2,3,6,7\}$, $K=8$) and the Singer set $C=\{0,1,3\}$ modulo $Q=7$ ($q=2$). Row $u$ is the block $\{uQ+1,\dots,uQ+Q\}$ of $[KQ]=[56]$; the shaded cells are the 18 elements of $A=\{uQ+c+1\}$. A sum $m$ determines the residues $\{c_0,d_0\}$ of its summands and their layer sum $T=(m-2-c_0-d_0)/Q$. The outlined cells are the summands in the four representations of $m=52$ (Example 3.4), one for each ordered layer pair $(u,v)$ with $u+v=7$, where $u$ is the layer of the summand with residue 0.

Fix $m$ with $R_A(m)>0$, and consider any representation $m=x+y$ with $x=uQ+c+1$, $y=vQ+d+1$ in $A$. Reducing modulo $Q$, $c+d\equiv m-2$. By Lemma 3.2 the multiset $\{c,d\}$ is therefore the same, say $\{c_0,d_0\}$, for every representation of $m$; consequently the integer $c+d=c_0+d_0$ is the same, and so is the layer sum

$$u+v=T:=\frac{m-2-c_0-d_0}{Q}.$$

*Case* $c_0\neq d_0$. In every representation $\{x,y\}$ exactly one summand has residue $c_0$. Map the representation to the pair (layer of the summand with residue $c_0$, layer of the summand with residue $d_0$). This is a pair $(u,v)\in P^2$ with $u+v=T$, and it determines both summands, so the map is injective. It is in fact a bijection, since every $(u,v)\in P^2$ with $u+v=T$ gives the representation $m=(uQ+c_0+1)+(vQ+d_0+1)$. Hence $R_A(m)=\rho_P(T)\le r$.

*Case* $c_0=d_0$. Now the representations are exactly the pairs $\{uQ+c_0+1,\ vQ+c_0+1\}$ with $u,v\in P$, $u\le v$, $u+v=T$, so $R_A(m)=\#\{(u,v)\in P^2 : u\le v,\ u+v=T\}\le\rho_P(T)\le r$. □

**Example 3.4.** Take $r=4$, the pattern $P=\{0,1,2,3,6,7\}$ of Lemma 3.6 below (so $K=8$), and the Singer set $C=\{0,1,3\}$ modulo $Q=7$; the lifted set $A$ is shown in Fig. 2. Let $m=52$. The residues of a representation $52=x+y$ satisfy $c+d\equiv 50\equiv 1 \pmod 7$, which for $c,d\in C$ forces $\{c,d\}=\{0,1\}$; hence $T=(52-2-0-1)/7=7$. The ordered pairs $(u,v)\in P^2$ with $u+v=7$ are $(0,7)$, $(1,6)$, $(6,1)$, $(7,0)$, and placing the summand with residue 0 in layer $u$ and the one with residue 1 in layer $v$ gives

$$52=1+51=8+44=43+9=50+2.$$

These are all the representations of 52 in $A$, so $R_A(52)=\rho_P(7)=4=r$. The layer pairs $(1,6)$ and $(6,1)$ give the different representations $8+44$ and $43+9$, which is why the pattern must be controlled by an ordered count.

The next corollary is the case $h=2$ of Plagne's formulation [Pla01, (10)–(11)]. Lemma 3.3 first gives sets in intervals of length $K(q^2+q+1)$; the prime number theorem lets us fit one into every sufficiently large interval with negligible loss, so that we obtain a bound for the lim inf and not only along a subsequence.

**Corollary 3.5.** *Let $K \geq 1$. If $P \subseteq \{0, 1, \ldots, K-1\}$ satisfies $\rho_P(t) \leq r$ for all $t$, then*

$$\liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}} \;\geq\; \frac{|P|}{\sqrt{K}}.$$

*Proof.* The empty pattern is immediate, so assume $P \neq \varnothing$. Fix $0 < \varepsilon < 1$ and put $X = \sqrt{N/K}$. By the prime number theorem, for all large $N$ there is a prime $q$ with $(1-\varepsilon)(X-1) < q \leq X-1$. Then $Q = q^2 + q + 1 \leq (X-1)^2 + (X-1) + 1 = X^2 - X + 1 \leq X^2$, so $KQ \leq N$. Applying Lemma 3.3 with a Singer set $C$ (Theorem 3.1) and using monotonicity,

$$S_r(N) \geq S_r(KQ) \geq |P|(q+1) > |P|\big((1-\varepsilon)(X-1)+1\big) \geq (1-\varepsilon)\frac{|P|}{\sqrt{K}}\sqrt{N}.$$

As $\varepsilon$ was arbitrary, the claim follows. □

3.3. **A two-block pattern.** It remains to choose a good pattern. An interval $P = \{0, \ldots, r-1\}$ has $\rho_P \leq r$ and gives only $|P|/\sqrt{K} = \sqrt{r}$, the same constant as for differences. The reason is that the ordered representation function of an interval is a tent, reaching height $r$ only at its central sum; there is room left in its tails. Cilleruelo, Ruzsa and Trujillo [CRT02, Lemma 2.3] fill part of the room in the right tail with $s = \lfloor r/2 \rfloor$ further layers $r+1, r+3, \ldots, r+2s-1$, spaced two apart. We use the following variant, in which the $s$ extra layers are consecutive and follow a gap of $s$ empty layers (Fig. 3). It has the same size and the same largest element as their pattern, so it gives the same constant; for $r \leq 3$ the two patterns coincide.

**Lemma 3.6** (a two-block variant of the pattern of [CRT02])**.** *Let $r \geq 1$, $s = \lfloor r/2 \rfloor$, and*

$$P_r = I \cup J, \qquad I = \{0, 1, \ldots, r-1\}, \quad J = \{r+s,\, r+s+1, \ldots, r+2s-1\}.$$

*Then $|P_r| = r + s$, $P_r \subseteq \{0, 1, \ldots, r+2s-1\}$, and $\rho_{P_r}(t) \leq r$ for all $t$.*

*Proof.* The first two claims are clear ($I$ and $J$ are disjoint as $r \leq r+s$). For finite sets $X, Y$ write $\rho_{X,Y}(t) = \#\{(x, y) \in X \times Y : x + y = t\}$; then

$$\rho_{P_r} = \rho_{I,I} + 2\rho_{I,J} + \rho_{J,J}.$$

The three terms are supported on $[0, 2r-2]$, $[r+s, 2r+2s-2]$ and $[2r+2s, 2r+4s-2]$ respectively, so the support of $\rho_{J,J}$ is disjoint from the other two. We have $\rho_{I,I}(t) \leq r$, $\rho_{J,J}(t) \leq |J| = s \leq r$ and $2\rho_{I,J}(t) \leq 2|J| = 2s \leq r$, which proves the claim for every $t$ outside the overlap $[r+s, 2r-2]$ of the first two supports.

This overlap is nonempty only if $s \leq r-2$; write its elements as $t = r + s + u$ with $0 \leq u \leq r-s-2$. Since $t \geq r-1$, the tent $\rho_{I,I}$ is on its descending side: $\rho_{I,I}(t) = 2r-1-t = r-s-1-u$. Next, $\rho_{I,J}(t)$ is the number of pairs $(x, j)$ with $x + j = u$, $0 \leq x \leq r-1$, $0 \leq j \leq s-1$ (put $y = r+s+j \in J$); since $u \leq r-1$, this is the number of $j \in [0, s-1]$ with $j \leq u$, namely $\min(u+1, s)$. Hence, using $2\min(a, b) \leq a + b$,

$$\rho_{P_r}(t) = (r-s-1-u) + 2\min(u+1, s) \leq (r-s-1-u) + (u+1) + s = r. \qquad \square$$

**Proposition 3.7** (Cilleruelo–Ruzsa–Trujillo [CRT02, Thm. 2.1])**.** *For every $r \geq 1$, with $s = \lfloor r/2 \rfloor$,*

$$\liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{N}} \;\geq\; \frac{r+s}{\sqrt{r+2s}}.$$

*Proof.* Apply Corollary 3.5 to the pattern $P_r$ of Lemma 3.6, with $K = r + 2s$. □

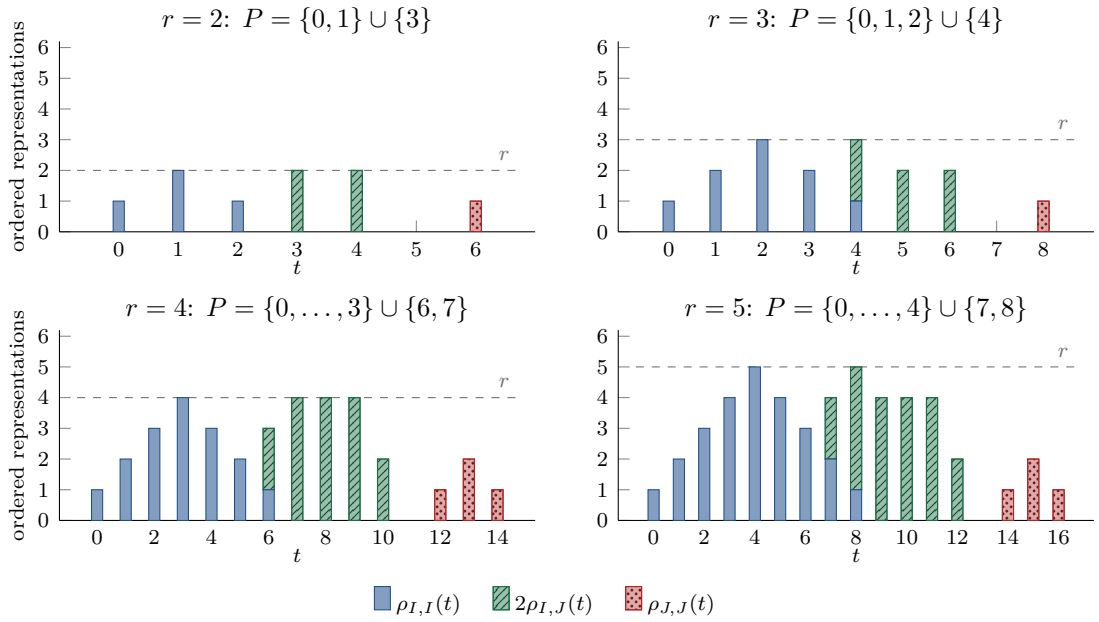



**Figure 3.** The ordered representation counts $\rho_{P_r}(t)$ of the pattern of Lemma 3.6 for $r = 2, 3, 4, 5$, split as $\rho_{I,I} + 2\rho_{I,J} + \rho_{J,J}$. The dashed line is height $r$. The cross terms $2\rho_{I,J}$ overlap only with the descending part of the tent $\rho_{I,I}$, where their combined count remains at most $r$, and $\rho_{J,J}$ lies beyond everything else. Each pattern attains the bound $r$.

*Remark* 3.8. Corollary 3.5 applies to any pattern. For $r = 2$ the condition $\rho_P \le 2$ says exactly that $P$ is a Sidon set, and the Sidon set $\{0, 1, 4, 6\}$ gives $4/\sqrt{7} \approx 1.5118$, better than the value $3/2$ from $P_2 = \{0, 1, 3\}$. This is the bound of Habsieger and Plagne [HP02], who also showed that no pattern satisfying $\rho_P \le 2$ gives a larger value of $|P|/\sqrt{K}$ in Corollary 3.5 [Pla01, §2.2]. Plagne [Pla01] found patterns improving on Proposition 3.7 for some other small $r$: for $r = 4$ the 12-element pattern $\{0, 1, 3, 9, 10, 11, 13, 18, 24, 25, 29, 30\}$ has $\rho_P \le 4$ and gives $12/\sqrt{31} \approx 2.1552$, and for $r = 6$ the pattern $\{0, 1, 2, 3, 4, 5, 8, 9, 12, 14, 18, 19\}$ gives $12/\sqrt{20} \approx 2.6832$ (we have checked both conditions $\rho_P \le r$ by computer).

### 3.4. **Proof of the main theorem.**

*Proof of Theorem 1.1.* The first inequality is Proposition 2.1, the last is Proposition 3.7, and the middle one is (1). □

For $r = 1$ the argument gives no separation, as expected: $s = 0$, $P_1 = \{0\}$, and the upper and lower bounds both equal 1, recovering $c_1 = c_1' = 1$.

## 4. What is known about the two constants

### 4.1. **Differences.** The constant $c_r'$ is known exactly.

**Theorem 4.1** ([CMT15, Thm. 3.10]; [ST25, Thm. 3])**.** *For every fixed $r \ge 1$, $D_r(N) = (1 + o(1))\sqrt{rN}$ as $N \to \infty$. Thus $c_r'$ exists and equals $\sqrt{r}$.*

The upper bound is Proposition 2.1. The lower bound comes from projecting a Bose set; we sketch the argument of [CMT15, ST25] for completeness. Let $q$ be a prime with $q \equiv 1 \pmod r$. Bose [Bos42] constructed a set $\mathcal{B} \subseteq \mathbb{Z}/(q^2 - 1)\mathbb{Z}$ of $q$ residues whose $q(q-1)$ nonzero differences are distinct and are precisely the residues not divisible by $q + 1$. Put $M = (q^2 - 1)/r$; since

$r \mid q-1$, $M$ is a multiple of $q+1$, so the kernel of the reduction map $\pi : \mathbb{Z}/(q^2-1)\mathbb{Z} \to \mathbb{Z}/M\mathbb{Z}$ consists of multiples of $q+1$. Hence $\pi$ is injective on $\mathcal{B}$ (no nonzero difference of $\mathcal{B}$ lies in the kernel), and for each nonzero $\delta \in \mathbb{Z}/M\mathbb{Z}$ the $r$ residues in $\pi^{-1}(\delta)$ are each a difference of $\mathcal{B}$ at most once, so $\delta$ has at most $r$ representations as a difference in $\pi(\mathcal{B})$. Taking representatives in $[M]$, every $d \in [1, M-1]$ is nonzero modulo $M$, so $D_r(M) \geq q = \sqrt{rM+1}$. Finally, given $N$, let $q$ be the largest prime with $q \equiv 1 \pmod r$ and $q \leq \sqrt{rN+1}$; then $M \leq N$, so $D_r(N) \geq D_r(M) \geq q$, and the prime number theorem in the progression $1 \bmod r$ gives $q \sim \sqrt{rN}$. Xu [Xu18] had earlier obtained results on popular differences from which the upper bound, and the limiting behaviour as $r \to \infty$, can be deduced [ST25].

4.2. **Sums.** In the language of [CRV10], a set $A \subseteq \mathbb{Z}$ with $R_A \leq r$ is the same as a set whose *ordered* representation function is at most $2r$ (in $\mathbb{Z}$ the ordered count of $m$ is $2R_A(m)$ or $2R_A(m)-1$). Thus $S_r(N) = \beta_{2r}(N)$ in [CRV10] and $S_r(N) = F_{2,r}(N)$ in [Pla01]; in [HP18] the notation is $F(r, N-1)$, since the ambient interval there is $\{0, \dots, N-1\}$. It is not known whether $S_r(N)/\sqrt{N}$ converges for any $r \geq 2$ [CRV10, Pla01]. The known bounds include:

- *Trivial upper bound.* The $\binom{|A|+1}{2}$ unordered pairs have sums in $[2, 2N]$, so $\binom{|A|+1}{2} \leq r(2N-1)$ and $\limsup S_r(N)/\sqrt{N} \leq 2\sqrt{r}$.
- *Better upper bounds* come from Fourier analysis; Habsieger and Plagne [HP18, Thm. 1] prove $\limsup S_r(N)/\sqrt{N} \leq \sqrt{1.740463\,(2r-1)}$, which improved the previous records for $2 \leq r \leq 5$; for $r = 2$ it gives 2.2851.
- *Lower bounds* for small $r$ come from glued constructions as in Lemma 3.3: Proposition 3.7 and the improvements in Remark 3.8. Habsieger and Plagne conjectured that $c_2 = 2$ [HP02]; see [Pla01, §4].
- *Large $r$.* Call $f$ *admissible* if it is a nonnegative integrable function supported on $[0,1]$ (extended by zero to $\mathbb{R}$), and write

$$(f * f)(x) = \int_{\mathbb{R}} f(t) f(x-t)\, dt.$$

Let $\sigma$ be the supremum of $\int_0^1 f$ over admissible $f$ with $f * f \leq 1$ everywhere. Cilleruelo, Ruzsa and Vinuesa [CRV10, Thm. 1.5] proved that

$$\lim_{r\to\infty} \liminf_{N\to\infty} \frac{S_r(N)}{\sqrt{2rN}} = \lim_{r\to\infty} \limsup_{N\to\infty} \frac{S_r(N)}{\sqrt{2rN}} = \sigma.$$

Since $c_r' = \sqrt{r}$, this says that both the lim inf and the lim sup of $S_r(N)/\sqrt{N}$, divided by $c_r'$, tend to $\sqrt{2}\,\sigma$ as $r \to \infty$. An explicit admissible function of Matolcsi and Vinuesa [MV10] gives $\sigma \geq 1.1509$. For the upper bound, Cloninger and Steinerberger [CS17] proved that $\sup(g * g) \geq 1.28\,(\int g)^2$ for every nonnegative integrable $g$ supported on $[-\frac14, \frac14]$; in their normalization 1.28 is a lower bound for $2/\sigma^2$. To check this, put $g(y) = f(2y + \frac12)$ for an admissible $f$: then $\int g = \frac12 \int f$ and $(g*g)(x) = \frac12 (f*f)(2x+1)$, so their inequality becomes $\sup(f*f) \geq 0.64\,(\int f)^2$, and $f * f \leq 1$ forces $\int f \leq 1/0.8$. Hence

$$1.1509 \leq \sigma \leq \tfrac54, \qquad 1.6276 < \sqrt{2}\,\sigma \leq \tfrac{5\sqrt{2}}{4} < 1.7678,$$

much more than the factor $\approx 1.06$ given by Theorem 1.1. Slightly better constants have been reported in recent preprints: $2/\sigma^2 \leq 1.5032$ from a computer-found function [GGTW25, Problem 6.2], and $2/\sigma^2 \geq 1.2802$ [IX26, Thm. 5], which would give $1.1534 < \sigma < 1.2500$.

**Table 1.** Constants supplied by the two-block construction. The second column is $c'_r = \sqrt{r}$ (Theorem 4.1), rounded to nearest. The third is the lower bound $(r+s)/\sqrt{r+2s}$ for $\liminf S_r(N)/\sqrt{N}$ from Proposition 3.7, and the fourth its ratio to $c'_r$; both are rounded down. The last column is the upper bound $\sqrt{1.740463(2r-1)}$ for $\limsup S_r(N)/\sqrt{N}$ from [HP18], rounded up. Better lower bounds are known for $r=2$ ($4/\sqrt{7} \approx 1.5118$ [HP02]) and $r=4$ ($12/\sqrt{31} \approx 2.1552$ [Pla01]); see Remark 3.8.

| $r$ | $c'_r$ | two-block lower bound | lower bound/$c'_r$ | HP18 upper bound |
|---|---|---|---|---|
| 2 | 1.4142 | 1.5000 | 1.0606 | 2.2851 |
| 3 | 1.7321 | 1.7888 | 1.0327 | 2.9500 |
| 4 | 2.0000 | 2.1213 | 1.0606 | 3.4905 |
| 5 | 2.2361 | 2.3333 | 1.0434 | 3.9578 |

## Acknowledgements and provenance

The observation that Problem #863 follows from the literature, and the proofs in the original note [Ho26], were found by GPT-5.4 Pro prompted by Boon Suan Ho, who announced them on the erdosproblems.com forum on 22 April 2026 [Thr]; Nat Sothanaphan reported on the same thread that a check of the note found no issues, and the problem page [Blo] was updated accordingly. The original note already used the two-block pattern of Lemma 3.6, describing the construction as a special case of that of [CRT02]. Relative to that note, this version adds the explicit error term in Proposition 2.1 (weaker than the one in [CMT15]), states the lifting lemma for arbitrary patterns as in [CRT02, Lemma 2.2], and adds the figures, Example 3.4, and the survey in Section 4. The papers [CMT15, ST25] on $r$-Golomb rulers, which already contain $c'_r = \sqrt{r}$, are not mentioned in the original note or on the problem page. This text was written by Claude Opus 5.5.

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